

Project Quick Run Guide

Hostel Roommate Compatibility & Smart Allocation System

1. Quick Commands to Run the Project

Open your terminal in the project root folder and execute the following commands:

Step 1: Compile

```
javac -d bin -sourcepath src src/Main.java
```

Step 2: Run

```
java -cp bin Main
```

2. System Login Credentials

Role / Access Level	ID / Email	Password
Administrator	admin (or admin@hostel.edu)	admin123
Student	ST104 (or ST106, ST108, ST101)	pass123

3. Recommended Step-by-Step Demo Flow

Follow this logical 4-step sequence to test all core functionalities effectively:

Step A: Multithreading & Synchronization Demo

From the Main Menu, select the **Run Concurrency Allocation Demo** option.

- Two student threads (Thread-Rahul and Thread-Karan) attempt to book the last available bed in Room B-203 at the exact same millisecond.

- **Outcome:** Demonstrate that exactly 1 student secures the bed, while the second student is safely rejected (throwing a `RoomFullException`), perfectly preventing race conditions and overbooking.

Step B: Admin Dashboard & Statistics

Login as Administrator (admin / admin123).

1. Select the **View System Statistics** option: Show the dynamically calculated stats, including total students, available beds, and compatibility distribution.
2. Select the **View All Rooms** option: Display the real-time room occupancy, block tracking, and bed status.
3. Select the **Logout** option to return to the main menu.

Step C: Student Compatibility & Match Finder

Login as Student (ST104 / pass123).

1. Select the **View Profile** option: Show the student's current profile and allocation status.
2. Select the **View Preferences** option: Display the 10 lifestyle preferences saved for this student.
3. Select the **Find Best Roommate Matches** option: Show how the system calculates compatibility scores across all eligible students and ranks the top matches (e.g., ST114 at 95%, ST106 at 92.5%).
4. Select the **View Compatibility Explanation** option and enter ST106.
 - Show the 10-category breakdown, the specific strengths of the match, any potential friction points, and the final system recommendation.

Step D: Smart Room Allocation

While still logged in as ST104.

1. Select the **View Room Allocation** option.
2. The system will detect that the student is unallocated and prompt: Would you like to trigger Smart Room Allocation now? (yes/no).
3. Type yes.
4. Demonstrate that the system automatically places the student into a highly compatible, partially-filled room (e.g., C-101 next to Neha Joshi).
5. Select the **View Profile** option to verify that the status has updated to **ALLOCATED (Room C-101 Bed 2)**.
6. Select the **Logout** option to conclude the demo.